

MATH 1342 – Elementary Statistics – Week 1: Introduction and the Structure of Data

Aug 24 – 26 | Introduction: What Is Statistics For? • 1.1 The Structure of Data | Fall 2026 • Section 019 • MW 7:30–8:50 PM • RRC Room 8210

Course-level Objectives	Module-level Objectives	Learning Materials	#	Activities	#	Assessments	#	
1. Interpret ideas of population versus sample, random variables, and techniques of descriptive statistics including frequency distributions, histograms, boxplots, and scatterplots. 2. Calculate and interpret measures of central tendency and dispersion, including mean, median, standard deviation, and quartiles. 7. Apply ideas of appropriate sampling techniques and experimental design to data production.	1. Describe statistics as the process of collecting, describing, and analyzing data, and judge a striking result by how likely it is to occur by chance alone. (CO1) 2. Distinguish a sample from a population and explain how a non-representative sample biases an estimate. (CO1, CO7) 3. Identify the cases and variables in a dataset presented as a table. (CO1) 4. Classify a variable as categorical or quantitative. (CO1) 5. Identify the explanatory and response variables in a research question. (CO1) 6. Compute and compare the mean and median of a small dataset and explain how an outlier affects each. (CO2)	START HERE: Welcome; Syllabus; Textbook and Wiley Course Resources; Online Tutoring; College Policies & Student Support Services	None	Participation: Syllabus – course tour and expectations	None	Online Homework: 1.1 The Structure of Data (due Sun Aug 30, 11:59 PM)	MO3, MO4, MO5	
		Introduction Slides: What Is Statistics For? (Day 1)	MO1, MO2, MO6	Day-1 class data activity: collect everyone's age, then find the mean, median, mode and boxplot together; compare the class median to the U.S. median age	MO1, MO2, MO6			
		e-Text: 1.1 The Structure of Data	MO3, MO4, MO5			Participation: Variables – identify cases, variables and variable types	MO3, MO4, MO5	
		Interactive Lecture: 1.1 The Structure of Data	MO3, MO4, MO5					
		Slides: 1.1 The Structure of Data	MO3, MO4, MO5					